Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 5

Title:Customer Analysis Dashboard

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GitHub Link: https://github.com/laishramritikumarsingh/BA-Assignment

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1. Introduction

The Customer Analysis Dashboard project uses Microsoft Power BI to analyze customer and order data. The report combines customer details with transaction information to provide an overview of sales activity, product-category popularity, payment preferences, monthly revenue patterns, and customer purchasing behaviour.

2. Objectives

- Identify the product category with the highest number of orders.
- Determine the most frequently used payment method.
- Analyze how revenue changes over time.
- Identify customers who place the most orders.
- Examine whether revenue is concentrated among a small group of customers.
- Present the findings through interactive and easy-to-read Power BI visuals.

3. Dataset Description

Two CSV datasets were used. The Customers dataset contains 1,000 customer records and demographic or segmentation attributes. The Orders dataset contains 5,000 order records and fields describing order date, product, category, quantity, unit price, discount, sales amount, and payment method.

3.1 Customer Dataset

Important fields include CustomerID, CustomerName, Gender, Age, City, State, IncomeBand, SignupDate, and CustomerSegment. CustomerID acts as the key used to connect customers to their orders.

CustomerID	CustomerName	Gender	Age	City	State	IncomeBand	SignupDate	CustomerSegment
C0002	Rahul Nair	Male	42	Mumbai	Maharashtra	Medium	27 July 2024	Low Value
C0004	Akash Iyer	Male	31	Delhi	Delhi	Medium	01 August 2024	Low Value
C0005	Pallavi Mehta	Female	29	Bengaluru	Karnataka	Medium	25 October 2023	Low Value
C0008	Varun Pillai	Male	32	Kolkata	West Bengal	Medium	01 February 2024	Low Value
C0010	Akash Iyer	Male	27	Mumbai	Maharashtra	Medium	21 October 2023	Low Value
C0011	Shreya Patel	Female	38	Mumbai	Maharashtra	Medium	27 October 2024	Low Value
C0013	Aditya Shetty	Male	33	Pune	Maharashtra	Medium	15 April 2024	Low Value
C0014	Ishita Das	Female	30	Ahmedabad	Gujarat	Medium	24 January 2023	Low Value
C0019	Kiran Chowdhury	Male	49	Ahmedabad	Gujarat	Medium	03 March 2023	Low Value
C0023	Neha Kulkarni	Female	36	Mysuru	Karnataka	Medium	09 October 2023	Low Value
C0025	Divya Das	Female	38	Kolkata	West Bengal	Medium	04 August 2023	Low Value
C0026	Vijay Chowdhury	Male	36	Bengaluru	Karnataka	Medium	30 June 2023	Low Value
C0027	Tanvi Patel	Female	29	Chennai	Tamil Nadu	Medium	01 December 2023	Low Value
C0030	Harish Das	Male	27	Delhi	Delhi	Medium	23 June 2023	Low Value
C0034	Meera Das	Female	27	Mumbai	Maharashtra	Medium	26 November 2024	Low Value
C0036	Pallavi Patel	Female	38	Bengaluru	Karnataka	Medium	08 August 2023	Low Value
C0039	Harish Verma	Male	43	Hyderabad	Telangana	Medium	09 June 2023	Low Value
C0043	Sneha Verma	Female	33	Delhi	Delhi	Medium	19 January 2024	Low Value
C0044	Deepika Verma	Female	29	Hyderabad	Telangana	Medium	12 August 2023	Low Value
C0045	Vijay Rao	Male	54	Ahmedabad	Gujarat	Medium	18 May 2023	Low Value
C0049	Swathi Kulkarni	Female	36	Delhi	Delhi	Medium	07 January 2025	Low Value
C0050	Arjun Bhat	Male	21	Bengaluru	Karnataka	Medium	26 January 2025	Low Value
C0055	Aarav Das	Male	40	Delhi	Delhi	Medium	04 January 2025	Low Value
C0057	Tanvi Iyer	Female	50	Mumbai	Maharashtra	Medium	24 March 2025	Low Value
C0062	Pallavi Pillai	Female	34	Chennai	Tamil Nadu	Medium	20 April 2023	Low Value
C0073	Amit Sharma	Male	20	Ahmedabad	Gujarat	Medium	07 April 2024	Low Value
C0079	Neha Rao	Female	30	Chennai	Tamil Nadu	Medium	06 April 2025	Low Value
C0080	Arjun Kulkarni	Male	35	Kolkata	West Bengal	Medium	03 October 2023	Low Value
C0081	Pranav Pillai	Male	41	Delhi	Delhi	Medium	31 January 2024	Low Value

customer_analysis_customers_1000 (1) (1,000 rows)

Figure 1. Customer dataset displayed in Power BI.

4. Data Model and Relationship

A one-to-many relationship was created between Customers[CustomerID] and Orders[CustomerID]. The Customers table is on the one side, and the Orders table is on the many side because one customer can place multiple orders. This relationship allows customer attributes to filter and summarize order data.

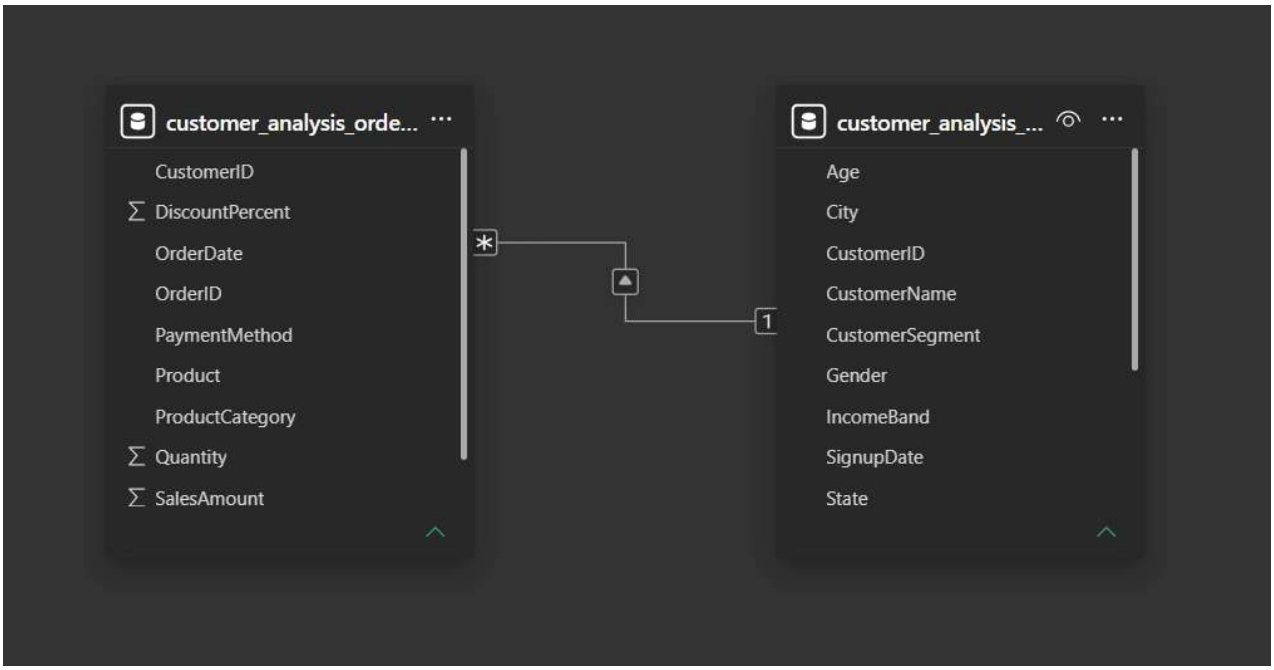


Figure 2. Power BI model showing the Customers-to-Orders relationship.

5. DAX Measures

Measures were created to calculate the key indicators used in the report. The measures are evaluated dynamically according to the filters and selections applied to the dashboard.

Total Orders

```
Total Orders = DISTINCTCOUNT('customer_analysis_orders_5000 (1)'[OrderID])
```

Counts unique order IDs.

Total Revenue

```
Total Revenue = SUM('customer_analysis_orders_5000 (1)'[SalesAmount])
```

Adds the sales amount for all orders.

Total Quantity

```
Total Quantity = SUM('customer_analysis_orders_5000 (1)'[Quantity])
```

Calculates the total quantity sold.

Average Order Value

```
Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)
```

Calculates average revenue per order.

Purchasing Customers

```
Purchasing Customers = DISTINCTCOUNT('customer_analysis_orders_5000 (1)'[CustomerID])
```

Counts distinct customers who placed at least one order.

```
Total Orders = DISTINCTCOUNT('customer_analysis_orders_5000 (1)'[OrderID])
```

Figure 3. DAX measure for Total Orders.

```
Total Quantity = SUM('customer_analysis_orders_5000 (1)'[Quantity])
```

Figure 4. DAX measure for Total Quantity.

```
Total Revenue = SUM('customer_analysis_orders_5000 (1)'[SalesAmount])
```

Figure 5. DAX measure for Total Revenue.

```
Average Order Value =
DIVIDE([Total Revenue], [Total Orders], 0)
```

Figure 6. DAX measure for Average Order Value.

```
Purchasing Customers =
DISTINCTCOUNT('customer_analysis_orders_5000 (1)'[CustomerID])
```

Figure 7. DAX measure for Purchasing Customers.

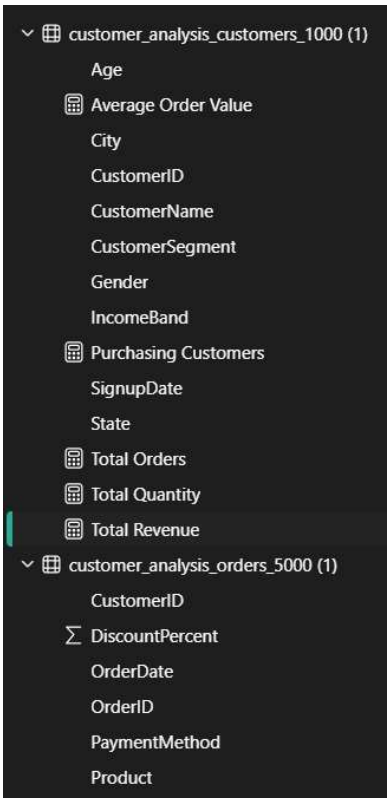


Figure 8. Power BI Fields pane showing the tables and measures.

6. Dashboard Design and Visualizations

The dashboard brings the main indicators together on a single report page. KPI cards provide a quick summary, while charts compare categories, payment methods, revenue over time, and customer-level activity.

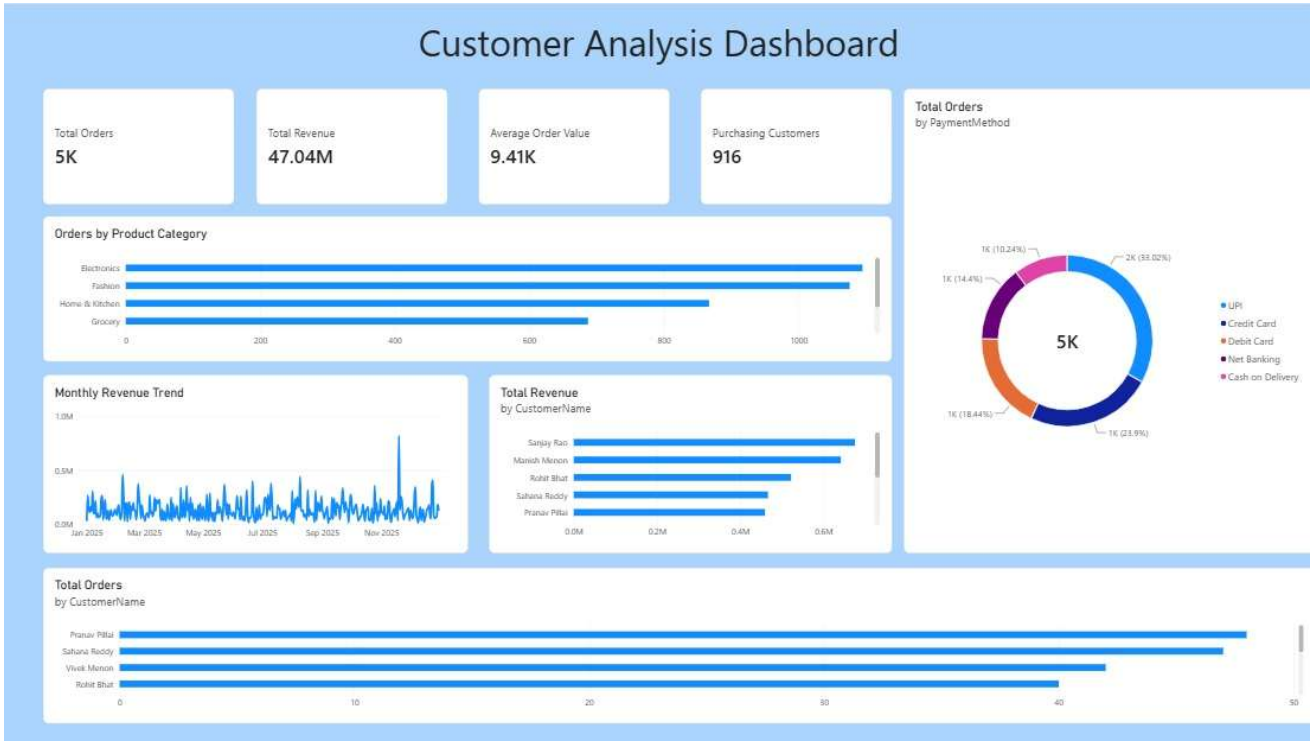


Figure 9. Completed Customer Analysis Dashboard in Power BI.

6.1 KPI Cards

Four cards were added: Total Orders, Total Revenue, Average Order Value, and Purchasing Customers. These cards provide a high-level summary of order volume, sales value, average transaction size, and the number of distinct customers who made purchases.

6.2 Orders by Product Category

A horizontal bar chart compares the number of orders across product categories. The category with the longest bar has the highest order count. This visual helps identify categories that are frequently purchased.

6.3 Orders by Payment Method

A donut chart displays the distribution of orders across payment methods. The size of each segment represents the number of orders made using that method, allowing payment preferences to be compared.

6.4 Monthly Revenue Trend

A line chart plots revenue against order date. For a month-by-month analysis, the date axis should be grouped by Year and Month rather than individual days. This makes changes in monthly revenue easier to interpret.

6.5 Customer Purchase and Revenue Analysis

Bar charts compare customers by order count and revenue. A Top N filter can be applied to display the ten customers with the highest order count or revenue. These views answer which customers purchase most often and which customers contribute the most sales value.

7. Key Findings

- **Total orders:** The dataset contains 5,000 orders.
- **Total revenue:** The dashboard displays approximately 47.04 million in sales amount.
- **Purchasing customers:** 916 distinct customers appear in the order data.
- **Payment method:** UPI is the most frequently used payment method in the provided dataset.
- **Category comparison:** The product-category bar chart compares order frequency across the six categories.
- **Customer concentration:** Customer revenue concentration can be assessed by comparing revenue from the top customers with overall revenue.

8. Conclusion

The Customer Analysis Dashboard demonstrates how Power BI can transform customer and order CSV files into a consolidated analytical report. The model relationship and DAX measures support the KPI cards and charts. The dashboard helps compare product categories, payment methods, revenue trends, and customer activity, providing a useful basis for business decisions.

9. Limitations and Further Improvements

The available order dataset includes SalesAmount but does not provide a dedicated cost or profit field. Therefore, actual profit cannot be calculated from the supplied fields alone. Future improvements could include cost data for profit analysis, slicers for date and customer segment, improved currency formatting, and a cumulative Pareto visual to quantify customer revenue concentration.